

Linear Algebra – Problem Set 1

Determinants, Cofactors & Adjugate, Cramer's Rule · Full solutions · Original problems in the spirit of MIT 18.06 (Strang)

Problem 1

A and B are 4×4 matrices with $\det A = 2$ and $\det B = -3$. Compute each of the following without finding any entries of A or B : (a) $\det(AB)$ (b) $\det(A^T)$ (c) $\det(A^{-1}B)$ (d) $\det(2A)$ (e) $\det(A^2B^{-1})$.

Problem 2

A is a 3×3 matrix with $\det A = -4$. A matrix B is obtained from A by performing, in order: $R_2 \leftarrow 3R_2$, then $R_1 \leftrightarrow R_2$, then $R_3 \leftarrow R_3 - 5R_1$. Find $\det B$ and explain the effect of each operation.

Problem 3

Starting from a 3×3 matrix A , the operations $R_2 \leftarrow 2R_2$, then $R_1 \leftrightarrow R_3$, then $R_3 \leftarrow R_3 + 4R_2$ are performed, producing

$$U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & -1 & 5 \\ 0 & 0 & 4 \end{bmatrix}.$$

Find $\det A$ without reconstructing A .

Problem 4

True or false? Justify each answer (if false, give a counterexample or correct the statement).

Problem 5

Let $A_t = \begin{bmatrix} t & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{bmatrix}$. Using determinant properties and row operations (not a full cofactor expansion), find $\det(A_t)$. For which values of t is A_t invertible?

Problem 6

Compute $\det \begin{bmatrix} 1 & 2 & 0 & 2 \\ 2 & 5 & 1 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & 3 & 1 & 3 \end{bmatrix}$ by row-replacement operations only (no swaps), reducing to upper triangular form.

Problem 7

Compute $\det \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 \\ 1 & 0 & 3 & 0 \\ 0 & 5 & 0 & 2 \end{bmatrix}$ by cofactor expansion, choosing the most efficient row or column and stating your choice.

Problem 8

Let $A = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$. Find the cofactors C_{21}, C_{22}, C_{23} , then compute $\det A$ by expanding along the second row. Explain the difference between a minor M_{ij} and a cofactor C_{ij} .

Problem 9

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$. (a) Find the cofactor matrix C and $\text{adj}(A)$. (b) Verify one complete row-by-column product in $A \text{adj}(A)$. (c) Find A^{-1} .

Problem 10

Use Cramer's Rule to solve $\begin{cases} 3x - 2y = 4 \\ 2x + 5y = 9 \end{cases}$. Identify A, A_1, A_2 and compute the necessary determinants.

Problem 11

Consider the system

$$\begin{cases} x + 2y + z = 5 \\ 2x - y + 3z = 4 \\ x + y + 2z = 7 \end{cases}$$

Use Cramer's Rule to find *only* y . Do not solve for x or z .

Problem 12

- (a) Suppose every row of a square matrix A sums to zero. Prove that $\det A = 0$ without expanding the determinant. (b) True or false: “ $\text{adj}(A)$ exists only when A is invertible.” Justify your answer.